Progressive unfreezing: the trainable suffix grows, the prefix stays frozen

24 transformer blocks. Black = frozen, red = trainable. Patch embedding, positional embedding and W_{proj} stay frozen; the linear head is always trained.

epoch 0: $k = 2$ blocks

8.29% trainable

epochs 12–13: $k = 8$

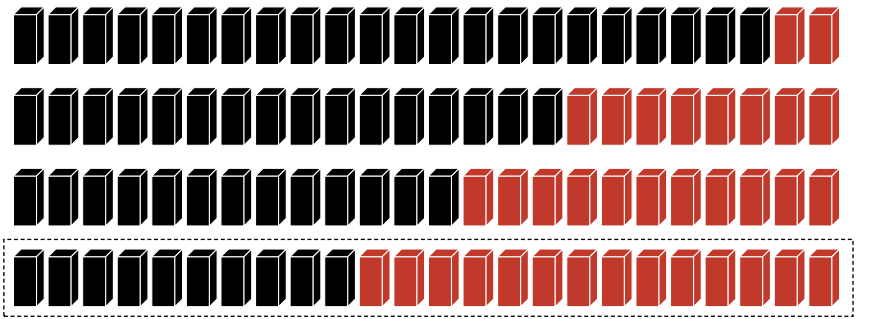
33.15% trainable

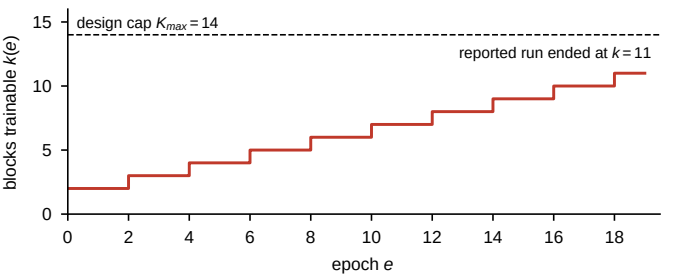
epochs 18–19: $k = 11$ (reported run)

45.58% trainable

design cap: $k = 14$ (not reached)

58.02% trainable





$$k(e) = \min(2 + \lfloor e/2 \rfloor,\ 14)$$